

APPLICATION NOTE

lifelines-hc: a Python package for HCHG testing in two-sample survival analysis

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Abstract

Motivation:

- Log-rank has good power against many alternatives, but Kipnis et al. [2026] showed it is *not* optimal against sparse hazard departures — concentrated differences in a few intervals at *unknown* locations.
- Weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) each pre-specify a temporal emphasis and are insensitive to sparse departures outside that window.

Results:

- **lifelines-hc**: Python package extending **lifelines** with the HCHG test (HC statistic applied to per-interval hypergeometric p-values; Kipnis et al. 2026).
- HCHG detects sparse hazard departures at unknown locations with no pre-committed temporal weight.
- Four real clinical datasets: CheckMate 057 PFS ($n = 582$), COMET-1 OS ($n = 1028$), AZURE DFS ($n = 3359$), and the CSL liver-cirrhosis trial ($n = 446$) — the last with fully public individual patient data.
- HCHG: $p \leq 0.014$ in all four. Log-rank non-significant throughout ($p \geq 0.26$); among the weighted tests only Gehan-Wilcoxon and Peto-Prentice on CheckMate 057 reach $p < 0.05$.

Availability and Implementation:

- <https://github.com/alonkipnis/lifelines-hc>; MIT licence.
- `pip install lifelines-hc`; requires Python ≥ 3.8 , **lifelines** ≥ 0.29 .
- Full analysis scripts and per-method results tables in Supplementary Data.

Key words: survival analysis; higher criticism; sparse signal detection; hypergeometric test; hypothesis testing; Python

Introduction

Background: log-rank and its limitation

- Kaplan-Meier [Kaplan and Meier, 1958] and log-rank [Mantel, 1966] are the standard tools for two-sample survival comparison.
- Schoenfeld [1981]: log-rank is asymptotically optimal against proportional-hazards (PH) alternatives.
- Kipnis et al. [2026]: log-rank is *not* optimal against **sparse** hazard departures — hazard differences confined to a small, *unknown* subset of time intervals.

Motivating biological examples of sparse departures

- *Crossing curves (immuno-oncology)*: PD-1 inhibitors require an immune-priming phase; hazard excess is confined

to early and late phases. CheckMate 057 (nivolumab vs docetaxel, NSCLC) [Borghaei et al., 2015].

- *Transient early benefit (targeted therapy)*: Cabozantinib (MET/VEGFR2 inhibitor) improved bone scan response at week 12 in COMET-1 (mCRPC) but the OS advantage was confined to the bone-response phase [Smith et al., 2016].
- *Subgroup-dependent accumulated benefit (pharmacology)*: Zoledronic acid's bone-microenvironment benefit is menopause-dependent; hazard difference accumulates only after premenopausal patients transition to postmenopause. AZURE trial [Coleman et al., 2011].
- *Time-varying treatment effect (hepatology)*: Prednisone in liver cirrhosis is harmful early (infection / fluid retention) but anti-inflammatory later; the survival curves cross more than once, with the effect concentrated in a few non-contiguous

windows. CSL trial [Martinussen and Scheike, 2006] — a canonical non-PH dataset with *fully public* individual patient data.

Existing non-PH alternatives and their limitation

- Weighted log-rank family: Gehan-Wilcoxon [Gehan, 1965], Peto-Prentice [Peto and Peto, 1972], Tarone-Ware [Tarone and Ware, 1977], Fleming-Harrington [Fleming and Harrington, 1991].
- Each applies a fixed temporal weight (early / mid / late emphasis).
- Requires knowing *a priori* where the hazard departure will occur — not suitable for the unknown-location sparse setting.
- All four can simultaneously miss a sparse departure if its location falls outside the intended window (demonstrated below).

The HCHG test and the present work

- Kipnis et al. [2026] proposed HCHG: the HC statistic of Donoho and Jin [2004] applied to per-interval HyperGeometric (HG) p-values from survival data.
- Designed specifically for sparse departures at unknown locations; no pre-specified temporal weight.
- No accessible Python implementation existed.
- **This paper:** `lifelines-hc` extends `lifelines` [Davidson-Pilon, 2019] with HCHG and supporting diagnostics.

The lifelines-hc package

The HCHG statistic

Step 1 — Hypergeometric interval p-values

- Partition follow-up into K equal-width intervals.
- Interval k : n_k total events, r_k^A/r_k^B at risk, x_k events in group B.
- Under the null of equal group-specific hazards:

$$x_k \mid n_k, r_k^A, r_k^B \sim \text{Hypergeometric}(r_k^A + r_k^B, r_k^B, n_k), \quad (1)$$

yielding a one-sided p-value p_k per interval.

Step 2 — HC aggregation (Donoho and Jin 2004)

- Order: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(K)}$.
- Apply the HC statistic to the ordered p-values:

$$\text{HCHG}_K = \max_{1 \leq j \leq \lfloor \gamma K \rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}(1 - p_{(j)})/K}}, \quad (2)$$

- $\gamma \in (0, 1)$, default 0.2: restricts scanning to the most extreme fraction of p-values [Donoho and Jin, 2004].
- Global p-value: label permutation ($n_{\text{perms}} = 2000$ in the analyses reported here).
- Two-sided: use min of two one-sided p_k in place of p_k .

Software design and API

Main functions

- `higher_criticism_test()` — HCHG test; returns `StatisticalResult` with `p_value` and `test_statistic`.
- `fisher_combination_test()` — Fisher combination ($-2 \sum_k \log p_k$) [Fisher, 1932].
- `min_p_test()` — Bonferroni-corrected MinP test.

- `suspected_deviations()` — per-interval HG p-values with `suspected` boolean column (HCHG-flagged intervals); used for diagnostic KM shading.
- `run_all_tests()` — benchmarks HCHG, Fisher, MinP, log-rank, and four weighted alternatives; returns `pandas.DataFrame`.
- Plotting: `plot_km_with_hc()`, `plot_pvalue_profile()`.

Usage example

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test,
                          ,
                          min_p_test,
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations
    =2000,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)           # permutation-
                                calibrated p-value
print(result.test_statistic)    # HCHG statistic
                                value
```

Design notes

- API consistent with `lifelines.statistics`.
- All three omnibus tests share the same HG interval p-values and permutation infrastructure.

Results

Overview

- Four real clinical datasets with distinct sparse non-PH patterns (Figure 1). Top row (A–D): KM curves; bottom row (E–H): per-interval p-value profiles.
- Benchmark: HCHG vs log-rank vs four weighted alternatives.
- Three datasets use IPD reconstructed from published KM curves; the CSL trial (Domain 4) has *fully public* raw IPD.
- Full per-method p-value tables in Supplementary Data.

Domain 1 — Immuno-oncology: crossing survival curves

Trial and data

- CheckMate 057 [Borghaei et al., 2015]: nivolumab ($n = 292$) vs docetaxel ($n = 290$), 2nd-line advanced non-squamous NSCLC.
- Endpoint: progression-free survival (PFS).
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Survival pattern

- Curves *cross*: docetaxel has an early advantage (immune priming), then nivolumab gains a durable late benefit.
- Sparse departure: elevated hazard for nivolumab confined to the early crossing window; then a late hazard reversal.

Test results

- Log-rank: $p = 0.351$ (NS) — early excess and late benefit cancel.

- Gehan-Wilcoxon: $p = 0.041$; Peto-Prentice: $p = 0.049$ (borderline) — their early emphasis happens to align with the crossing; not a general signal.
- Tarone-Ware: $p = 0.358$; Fleming-Harrington (1,1): $p = 0.256$ (both NS).
- HCHG: $p = 0.001$ — flags both the early and late divergence windows independently (Figure 1A,E).

Domain 2 — Targeted therapy: transient early benefit

Trial and data

- COMET-1 [Smith et al., 2016]: cabozantinib ($n = 682$) vs prednisone ($n = 346$), chemotherapy-pretreated mCRPC. 2:1 randomisation.
- Endpoint: overall survival (OS). Median OS ≈ 10 months.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Biological mechanism and survival pattern

- Cabozantinib inhibits MET and VEGFR2: reduces bone lesion activity (bone scan response at week 12 significantly improved).
- Resistance through alternative pathways develops rapidly; benefit is confined to the bone-response phase (months 3–7).
- Sparse departure: early OS advantage, then curves converge.

Test results

- Log-rank: $p = 0.262$ (NS).
- All four weighted alternatives: $p \geq 0.19$ (NS) — early-weighted tests (GW, PP) partially align but are diluted by the post-response null period.
- HCHG: $p = 0.014$; Fisher combination: $p = 0.028$ (Figure 1B,F).

Domain 3 — Adjuvant bisphosphonate: menopause-dependent effect

Trial and data

- AZURE [Coleman et al., 2011, 2014]: zoledronic acid ($n = 1681$) vs control ($n = 1678$), early-stage breast cancer. 10-year follow-up.
- Endpoint: disease-free survival (DFS). 973 events total.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

Biological mechanism and survival pattern

- Zoledronic acid targets the bone microenvironment (osteoclast inhibition), preventing micrometastasis in the bone niche.
- Benefit is menopause-dependent: requires low oestrogen / high bone resorption (postmenopausal state).
- Many patients were premenopausal at enrolment; benefit accumulates only after their transition to postmenopause (months 20–60).
- Sparse departure: hazard difference distributed across a mid-to-late window, null outside it.

Test results

- Log-rank: $p = 0.305$ (NS).
- All four weighted alternatives: $p \geq 0.28$ (NS) — no fixed weight captures the mid-to-late window unaffected by the null early period.
- HCHG: $p = 0.005$; Fisher: $p = 0.004$ (Figure 1C,G).

- Validated by published subgroup analyses: significant DFS benefit in postmenopausal patients [Coleman et al., 2011, 2014].

Domain 4 — Corticosteroid in cirrhosis: time-varying effect (public IPD)

Trial and data

- CSL trial [Martinussen and Scheike, 2006]: prednisone ($n = 226$) vs placebo ($n = 220$), histologically verified liver cirrhosis (Copenhagen, 1962–69). 270 deaths.
- Endpoint: overall survival. Follow-up up to ~ 13 years.
- **Raw IPD genuinely public** — distributed in the CRAN `timereg` package and the open `SurvSet` repository [Drysdale, 2022]; *not* reconstructed from a KM curve.
- Note: source CSV is counting-process (long) format with a time-varying prothrombin covariate; collapsed to one row per patient (final follow-up time and status) for the two-sample test.

Biological mechanism and survival pattern

- Corticosteroids have a time-varying effect in cirrhosis: early harm (immunosuppression, infection, fluid retention) followed by later anti-inflammatory benefit in actively inflamed subgroups.
- Survival curves cross more than once; treatment effect concentrated in a few non-contiguous windows (sign-changing).
- Canonical non-PH dataset, widely used to illustrate time-varying-coefficient models.

Test results

- Log-rank: $p = 0.384$ (NS).
- All four weighted alternatives: $p \geq 0.13$ (NS) — Gehan-Wilcoxon 0.51, Tarone-Ware 0.38, Peto-Prentice 0.45, Fleming-Harrington (1,1) 0.14.
- HCHG: $p = 0.002$ (significant); Fisher: $p = 0.034$ (Figure 1D,H).
- Demonstrates HCHG on genuinely public raw data, with a multi-window sign-changing departure no single weight can capture.

Cross-domain summary

- HCHG achieves $p \leq 0.014$ in all four domains; log-rank is non-significant throughout ($p \geq 0.26$).
- GW ($p = 0.041$) and PP ($p = 0.049$) reach $p < 0.05$ in CheckMate 057 only — the sole weighted results below 0.05; elsewhere all weighted tests $p \geq 0.13$. Their early emphasis coincidentally aligns with that trial's crossing signal and does not generalise.
- Fisher combination also significant in all four, consistent with sparse localised signals. MinP significant in CheckMate 057 and AZURE.
- The four patterns differ (early+late crossing, transient-early, mid-to-late accumulated, multi-window sign-changing), yet HCHG detects all without any per-dataset tuning.

Conclusion

- `lifelines-hc` provides a Python implementation of HCHG, the test specifically designed for sparse hazard departures at unknown temporal locations.
- Unlike weighted log-rank tests, HCHG requires no a priori specification of where the hazard departure occurs.
- Diagnostic output (`suspected_deviations()`) identifies which time windows drive the signal.
- Fisher combination and MinP provided as complementary omnibus alternatives sharing the same hypergeometric infrastructure.
- Practical complement to standard survival testing whenever the proportional-hazards assumption is uncertain.

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TODO.

Conflict of Interest

None declared.

Funding

TODO.

Data availability

- IPD for CheckMate 057, COMET-1, and AZURE reconstructed from published KM curves via Guyot et al. [2012] as implemented in the `kmdata` R package (<https://github.com/raredd/kmdata>).
- CSL liver-cirrhosis IPD is fully public: CRAN `timereg` package (dataset `cs1`) and the `SurvSet` repository [Drysedale, 2022] (<https://github.com/ErikinBC/SurvSet>).
- All analysis code: <https://github.com/alonkipnis/lifelines-hc/tree/main/AppNote>.

References

- Hossein Borghaei, Luis Paz-Ares, Leora Horn, David R. Spigel, Martin Steins, Neal E. Ready, Laura Q. Chow, Everett E. Vokes, Enriqueta Felip, Esther Holgado, et al. Nivolumab versus docetaxel in advanced nonsquamous non-small-cell lung cancer. *New England Journal of Medicine*, 373(17): 1627–1639, 2015. doi: 10.1056/NEJMoa1507643.
- Robert E. Coleman, Howard Marshall, David Cameron, David Dodwell, Robert Burkinshaw, Mary Keane, Michael Gil, Stephen J. Houston, Robert J. Grieve, Peter J. Barrett-Lee, et al. Breast-cancer adjuvant therapy with zoledronic acid. *New England Journal of Medicine*, 365(15):1396–1405, 2011. doi: 10.1056/NEJMoa1105195.
- Robert E. Coleman, Marie Collinson, Eleanor Banks, Graham Clack, Dorothy Ritchie, Victoria Liversedge, David Cameron, David Dodwell, Michael Jones, Peter Barrett-Lee, et al. Benefits and risks of adjuvant treatment with zoledronic acid in stage II/III breast cancer. 10 year follow-up of the AZURE randomized clinical trial. *Journal of Bone Oncology*, 3(2): 23–30, 2014. doi: 10.1016/j.jbo.2013.12.001.
- Cameron Davidson-Pilon. `lifelines`: survival analysis in Python. *Journal of Open Source Software*, 4(40):1317, 2019. doi: 10.21105/joss.01317.
- David Donoho and Jiashun Jin. Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics*, 32(3):962–994, 2004. doi: 10.1214/009053604000000265.
- Erik Drysdale. `SurvSet`: An open-source time-to-event dataset repository. *arXiv preprint arXiv:2203.03094*, 2022. doi: 10.48550/arXiv.2203.03094.
- Ronald A. Fisher. *Statistical Methods for Research Workers*. Oliver and Boyd, Edinburgh, 4th edition, 1932.
- Thomas R. Fleming and David P. Harrington. *Counting Processes and Survival Analysis*. Wiley, New York, 1991. ISBN 0-471-52218-X.
- Edmund A. Gehan. A generalized Wilcoxon test for comparing arbitrarily singly-censored samples. *Biometrika*, 52(1–2): 203–223, 1965. doi: 10.1093/biomet/52.1-2.203.
- Patricia Guyot, A. E. Ades, Mario J. N. M. Ouwers, and Nicky J. Welton. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan-Meier survival curves. *BMC Medical Research Methodology*, 12:9, 2012. doi: 10.1186/1471-2288-12-9.
- Edward L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958. doi: 10.1080/01621459.1958.10501452.
- A Kipnis, B Galili, and Z Yakhini. Higher criticism for rare and weak non-proportional hazard deviations in survival analysis. *Biometrika*, 113(1):asaf075, 02 2026. ISSN 1464-3510. doi: 10.1093/biomet/asaf075. URL <https://doi.org/10.1093/biomet/asaf075>.
- Nathan Mantel. Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemotherapy Reports*, 50(3):163–170, 1966.
- Torben Martinussen and Thomas H. Scheike. *Dynamic Regression Models for Survival Data*. Springer, New York, 2006. doi: 10.1007/0-387-33960-4. CSL liver cirrhosis trial (prednisone vs placebo, Copenhagen Study group for Liver diseases, 1962–1969); dataset `cs1` in the CRAN `timereg` package.
- Richard Peto and Julian Peto. Asymptotically efficient rank invariant test procedures. *Journal of the Royal Statistical Society, Series A*, 135(2):185–207, 1972. doi: 10.2307/

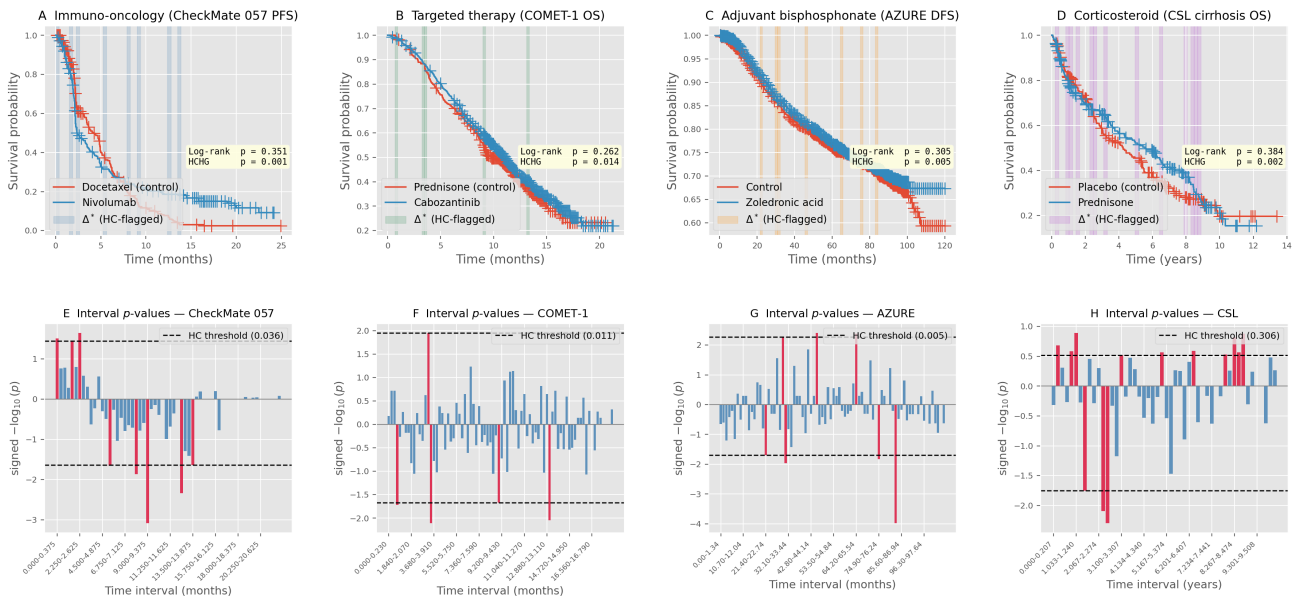


Fig. 1. HCHG detects non-proportional hazard departures across four clinical datasets with distinct temporal patterns. Top row (A–D): Kaplan-Meier curves with HCHG-flagged intervals (shaded). Bottom row (E–H): per-interval $\text{signed } -\log_{10}(\text{hypergeometric } p\text{-value})$ — upward bars mark excess events in the treatment arm, downward bars excess in the control arm. Dashed lines are the per-direction HCHG thresholds; bars reaching beyond either line are highlighted in red and coincide exactly with the intervals shaded above. (A,E) CheckMate 057 PFS (nivolumab vs. docetaxel, $n = 582$); log-rank $p = 0.351$, HCHG $p = 0.001$. Curves cross at ~ 3 months before nivolumab gains a durable late advantage. (B,F) COMET-1 OS (cabozantinib vs. prednisone, $n = 1028$); log-rank $p = 0.262$, HCHG $p = 0.014$. Early OS advantage (months 3–7) during bone-lesion response, fading as resistance develops. (C,G) AZURE DFS (zoledronic acid vs. control, $n = 3359$); log-rank $p = 0.305$, HCHG $p = 0.005$. Menopause-dependent benefit accumulates in months 20–60. (D,H) CSL OS (prednisone vs. placebo, liver cirrhosis, $n = 446$); log-rank $p = 0.384$, HCHG $p = 0.002$. Time-varying corticosteroid effect with multiple curve crossings; *fully public* raw IPD (CRAN `timereg` / `SurvSet`). In every dataset the log-rank test is non-significant while HCHG is significant. For the first three, IPD reconstructed via Guyot et al. [2012].

2344317.

David Schoenfeld. The asymptotic properties of nonparametric tests for comparing survival distributions. *Biometrika*, 68(1): 316–319, 1981. doi: 10.1093/biomet/68.1.316.

Matthew R. Smith, Christopher J. Sweeney, Paul G. Corn, Dana E. Rathkopf, David C. Smith, Maha Hussain, Daniel J. George, Celestia S. Higano, Andrea L. Harzstark, A. Oliver Sartor, et al. Cabozantinib in chemotherapy-pretreated

metastatic castration-resistant prostate cancer: results of a phase III randomized controlled trial (COMET-1). *Journal of Clinical Oncology*, 34(25):3005–3013, 2016. doi: 10.1200/JCO.2015.65.5597.

Robert E. Tarone and James Ware. On distribution-free tests for equality of survival distributions. *Biometrika*, 64(1):156–160, 1977. doi: 10.1093/biomet/64.1.156.